Evaluating Overvalued vs Undervalued F1 Drivers of the 2025 Formula 1 season

Aryavir Gulati
Institut Le Rosey, Château du Rosey
1180 Rolle, Switzerland
gulatiaryavir@gmail.com

Abstract

In F1, driver skill plays only a minor role in the results that they achieve on track; car performance plays a much bigger part, and driver salaries vary significantly from under $1 million to over $60 million per year. This study compares driver salary to on-track performance for the entire 2025 season (24 races + 6 sprint races, N = 21 drivers).On-track performance was defined as head-to-head win percentage against a teammate from the same team, as teammates use the same equipment. To reduce the impact on the model of the few outliers with high salaries, data were analyzed with robust regression in Stata. The results showed that salary was a statistically significant predictor of head-to-head performance (b = 1.09, p = .009), while age was not a significant predictor (p = .321).These results indicate that the earnings of a driver are significantly correlated with his or her on-track performance as compared to his teammate and that the drivers most over- or undervalued are the high-salary outliers.

Keywords: Formula 1, driver salary, performance evaluation, robust regression, sports economics

Introduction

In most sports, the result of the race is dependent on the driver more than the outcome of a Formula 1 race. In sports, the standard equipment is used that is independent of the skill of the driver, whereas an F1 driver's performance depends more on the car he or she is contracted to drive. Two drivers of the same level of skill can thus yield vastly different results due to their different cars. This is difficult to measure the quality of the drivers based on the raw results of the championships (points won or finishing position). Similarly, salaries in Formula 1 are also highly skewed, with rookie and reserve drivers earning less than $1 million annually, while more seasoned drivers make more than $60 million (Forbes, 2025). It is not uncommon to hear a lot of discussion about driver value in motorsport, but such discussions are rarely based on a methodology that seeks to distinguish between the skill of the driver and the quality of the car, and they are not typically statistically evaluated using formal methods. In most single-sport sports, such as tennis and golf, the equipment used is either standard or doesn't significantly affect an athlete's performance and prize money and endorsement value is more directly linked to an athlete's performance (Kahn, 2000). At the opposite end of the scale, in Formula 1 the car that a driver drives is as important as the driver's skill in winning a race, if not more so, for in some years the car is developed and built by the driver's team. This makes F1 an unusual sports economics research case, as the athlete-equipment relationship is more akin to motorsports than individual sports and the traditional performance metrics are less relevant as a stand-alone measure of individual performance. The core problem in this research is a problem of measurement: How to measure the performance of a particular driver in a sport where the equipment plays a major part of the game? The correlation between championship points and salary is difficult to understand – it is not a correlation between the driver's skill and competitiveness of his car. This can make it appear as though the low-paid driver is doing worse than the high-paid driver, even though the low-paid driver might actually be a better driver, if they are in a less competitive car, and the high-paid driver is in a more competitive car. Without

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being able to isolate driver ability from the car, it's hard to make any conclusive claims about whether or not F1 drivers are fairly paid, overvalued or undervalued. This is compounded by the fact that there are few drivers around. There are only 21 drivers who race in Formula 1 in any season, which means that there are fewer observations in an analysis and that a handful of extreme observations (such as a handful of extremely high-paid drivers) can have an impact on a model. With a small sample size and a very skewed distribution of wins, a detailed study of pay and performance in F1 has to take both of these factors into consideration.

It is important for the teams, drivers and fans to see if driver salary is a true reflection of on-track performance. For teams, it provides a direct link to how well resources are being allocated, as much of a team's budget is typically devoted to driver salaries, and it's crucial to determine whether the money is worth it or not, as it has tangible financial consequences. Raw statistics are not the best basis for salary negotiations for drivers and their management because they are easily confused by the performance of the equipment. This study offers an alternative to the wide-spread informal and opinionated driver value conversations found in motorsport commentary and not well defensible. The broader implications of this study are on the research methods used in sports economics to study equipment dependent sports disciplines in general. The vast bulk of the pay-performance research literature was developed in sports where the output of the individual is not mixed with the output of the equipment, e.g., basketball and baseball. The methodological tools developed for equipment neutral sports are becoming less relevant as the sport becomes increasingly commercialised and paid for by athletes with varying amount of pay, such as motorsport, cycling and sailing (Simmons, 2022). One way to do this more broadly and economically relevant sport is to use the car controlled, teammate-comparison method described in this study.

This research is aimed at determining whether driver salary is a good predictor of on-track performance, once statistically controlling for the quality of the car, and whether a linear regression is adequate to reveal the relationship between the two in the small sample of drivers in Formula 1, where there is a strong imbalance in the sample size.

The aim of this study is to see if the salary of an F1 driver can be used to predict their on-track performance, whilst holding car quality constant. The head-to-head win percentage is determined against a teammate on the same team during the entire 2025 season; any difference in win percentage can be explained mainly by driver skill since the teams are using the same equipment. The salary, age and years in F1 are tested as predictors; standard and robust regression techniques are used to address the well-known problems of skewed salary distributions and high-leverage outliers for small samples.

Literature Review

The connection between athlete pay and performance has long been a topic of discussion in sports economics. The seminal paper on pay and performance in Major League Baseball is by Scully (1974), who estimated the marginal revenue product of players, and determined that wages in the reserve clause were a mere fraction of the value of players to their teams. This method of tying individual performance measures to team results and then to team pay, provided the foundation for many decades of subsequent sports labor economics research (Humphreys & Pyun, 2015; Zimbalist, 1992).

Scully's MRP model, which attempts to measure a player's economic value to a team's success by correlating his team performance statistics with the team's winning percentage and then his attendance and team revenue, was the foundation of sports labor economics for decades after it was published (Kahn, 2000). Following this work, Zimbalist (1992) updated and improved the Scully model, noting some

methodological sensitivities with estimating player contribution, and warning that estimates of MRP should not be more precise than the data from which they are derived. This methodological stream is important to this study because it highlights a common thread in the literature, namely that pay-performance relationships are extremely dependent on the operationalization of performance itself, a consideration which is the motivation behind using a car-controlled performance measure, instead of raw statistics, in this study.

In formula 1, it is harder to separate the driver from the car and team, as opposed to, say, baseball. A review of the motorsport labour economics highlights the significant difference in driver wages based on team budget and driver experience, as well as the fact that there is a strong complementarity between driver skill and car tech, which makes any “pay-perform” correlation complex (IJFMR, 2024). Research relating to F1 has focused on the impact of team budgets, staff and financial investment on team performance, with the consensus being that financial resources are highly correlated with competitive outcomes at the team level (Duggal, 2021; Research Archive of Rising Scholars, 2024). Much of the literature, however, has been concerned with team expenditures and not driver wages, and has not usually accounted for the skill of the driver alone in assessing the pay-performance relationship, instead separating driver skill from car quality.

A complementary theoretical perspective is that of Rosen (1981), the economics of superstars. In markets where consumers value the best performers highly, and where the best performer can produce a product that many more people than usual can enjoy at little additional cost, differences in talent correspond to disproportionate differences in earnings, Rosen argued. In Formula 1, a couple of top-tier guys get a lot more media and commercial interest than the average driver on the grid, so it's a good bet that exactly that's what happens in Formula 1. On the other hand, if Rosen's formula holds true for F1, it would imply that the top earners are paid much more than a linear extrapolation of the middle of the grid's earnings and performance would suggest, which is broadly consistent with the very highest pay outliers of high-leverage performance levels identified later in this study in the diagnostic testing.

A more relevant methodological precedent is that of Humphreys and Pyun (2015), who documented monopsony exploitation in Major League Baseball position players under the league's reserve clause, in which players' salaries were systematically lower during their first few years of their careers when their pay is restricted by the reserve clause. As the results of this finding highlight, there are several points in the present study that are directly relevant to observed pay in professional sport that, for institutional and structural reasons, often deviates from what is predicted by performance alone, even in the absence of measurement error (Kahn, 2000). This should be noted as a significant limitation for the present analysis, since even the most restricted measure of performance (the car) is not necessarily the sole factor determining driver compensation, and the conclusions of the analysis are to be taken as evidence of association, and not as a comprehensive statement of the extent to which the F1 driver compensation system is responsive to the car.

The present study contributes to this literature by employing a teammate-comparison design, which is common in informal F1 analysis, but has not been applied in a formal statistical setting, in order to separate out the skill component from the car performance, and by testing to determine whether a standard linear regression is sufficient to identify a pay-performance relationship in a sport where the disparity in pay is extreme.

In addition to the theoretical framing, the general sports labor economics literature has also evolved a variety of empirical techniques to handle the endogeneity issues associated with pay-performance work, such as the fixed-effects panel models that capitalise on within-player variation across multiple seasons, and

instrumental variable models that aim to separate the effects of performance from the effects of pay, as outlined by Simmons (2022). These approaches are not possible in the single-season, cross-sectional design of the present study, as they are found in more mature literature on baseball and basketball. This gap is itself informative—that is, it gives information about the practical constraints of studying a sport with only twenty-one players per season—and it encourages an alternative approach to this study, which is based on robust regression instead of the panel or instrumental methods that are not feasible in this setting.

Methodology

The data were collected from all 24 Grand Prix races and 6 sprint races held during the 2025 season of Formula 1, which included all 21 drivers who participated in the season. The total points, average finishing position and head-to-head results between the driver and their teammate from the same team in each race were calculated for each driver, with the exception of races in which they both were not classified as finishers. The primary dependent variable was head-to-head win percentage – the percentage of races a driver would win if they were to race a race in which they were classified next to another teammate in the same car.

Teammate comparison is used here as a proxy for car quality rather than a complete isolation of driver skill: teammates generally share the same chassis and power unit, but may still differ in upgrade timing, setup choices, reliability, team strategy, and race incidents, any of which could contribute to a within-team performance gap independent of driver ability.

Driver salaries are based on published 2025 base salaries (RacingNews365, 2025) in millions of US dollars. Secondary independent variables were age and F1 experience (number of seasons in which a driver entered a race, including 2025). Due to missing salary data, one driver (Colapinto) was not included in salary-based analyses, leaving N = 20 for regression analyses.

This figure reflects base salary only, as reported by RacingNews365 in 2025, and excludes performance bonuses, image rights, and personal sponsorship income, which are not publicly disclosed for individual drivers. Because these figures are journalistic estimates rather than verified contractual data, they may contain measurement error that could not be corrected for in this analysis.

Analyses were done in Stata. Ordinary Least squares regression (OLS) was conducted with a dependent variable of head-to-head win percentage and the independent variables of salary, age and F1 experience. The variance inflation factors (VIF) were then calculated to determine multicollinearity, the Breusch-Pagan/Cook-Weisberg test was used to determine if there was heteroskedasticity, and the Shapiro-Wilk test was used to determine if the residuals were normally distributed. In order to minimize the influence of a few observations with higher salaries without dropping them from the sample, a robust regression (iteratively reweighted least squares with Huber and biweight weights) was also performed.

Stata's robust regression procedure first screens for gross outliers using Cook's distance, excluding any observation exceeding a threshold of 1 before the Huber and biweight reweighting stages are applied. In this analysis, one additional observation (Hamilton) was excluded at this initial screening step, reducing the robust-regression sample from N = 20 to N = 19. This exclusion, and its effect on the results, is discussed further in the Results and Limitations sections.

Robust regression was selected as the primary specification after comparing it against three alternative approaches to the same non-significant OLS result. Adding heteroskedasticity-robust standard errors to the OLS model did not change the conclusion (salary remained non-significant, $p = .302$). Log-transforming

salary before applying heteroskedasticity-robust standard errors produced a significant salary effect ($p = .023$), consistent with the robust-regression result. Robust regression was retained as the primary specification because, unlike a log transformation, it does not alter the interpretation of the salary coefficient, and because it explicitly identifies and reports which observations were down-weighted or excluded, providing more transparency about how the highest-salary cases affected the result than a transformation-based approach would.

Results

Table 1 shows descriptive statistics for the key study variables. The salary was widely dispersed (M = \$13.73 million, SD = \$18.72 million) and this indicates that there is significant variation in salary across the grid.

Table 1

Descriptive Statistics for Key Study Variables

Variable	n	M	SD	Min	Max
Salary (USD millions)	20	13.73	18.72	0.75	65.00
Age (2025)	21	27.29	6.60	19	44
F1 experience (seasons)	21	7.14	5.74	1	22
H2H win percentage	21	47.82	28.13	0	100

In standard OLS regression, none of the predictors tested were statistically significant. However, salary did not achieve significance ($p = .104$) and was not significant with age ($p = .194$) or F1 experience ($p = .217$). The overall model with all three predictors was also non-significant ($F(3,16) = 0.98$, $p = .426$); the adjusted R^2 was -.003, meaning that the model accounted for essentially none of the variance in performance.

There were two violations of the OLS assumptions, as suggested by the diagnostic test. There was a high level of multicollinearity with the variables age and F1 experience (VIF = 16.25 and 19.67, respectively), which is reflected in their strong bivariate correlation ($r = .962$, $p < .001$). The Breusch-Pagan/Cook-Weisberg test revealed that there was significant heteroskedasticity ($\chi2(1) = 4.56$, $p = .033$). Residuals were found to be normally distributed (Shapiro-Wilk $W = .933$, $p = .173$).

With these diagnostics and it being known that two of the observations (Verstappen and Hamilton) were extreme values in terms of the salary, a robust regression was carried out with salary and age as predictors, excluding F1 experience as it was collinear with age. In this model, head-to-head win percentage was, on average, 1.09 percentage points higher for each additional million dollars in salary (SE = 0.37, $p = .009$), while age was not a significant predictor ($b = 0.96$, $p = .321$). The full regression results are shown in Table 2.

Age was retained over F1 experience because age is measured with certainty for every driver, whereas the experience variable (seasons started) is more sensitive to how partial or reserve seasons are counted; age and experience were never included in the same model given their near-collinearity ($r = .962$), and the two were not treated as interchangeable substitutes for one another.

This N = 19 reflects one further exclusion beyond the N = 20 salary-based sample described in the Methodology: Hamilton's observation exceeded the Cook's distance threshold of 1 used by the robust regression procedure to screen for gross outliers prior to Huber and biweight reweighting, and was therefore excluded from this specific model. Because Hamilton represents a high-salary, low-performance case, this exclusion is itself relevant to the paper's central question and is addressed directly below.

Table 2

Robust Regression Predicting Head-to-Head Win Percentage

Predictor	b	SE	t	p
Salary (USD millions)	1.09	0.37	2.97	.009
Age	0.96	0.94	1.02	.321
Constant	12.36	24.87	0.50	.626

Note. N = 19. F(2, 16) = 6.51, p = .009.

To directly operationalize the paper's central over/undervaluation question, a driver-level residual was calculated from the robust regression model: the difference between each driver's actual head-to-head win percentage and the value predicted for them by the model given their salary and age. A positive residual indicates a driver outperforming what their salary and age would predict (undervalued); a negative residual indicates a driver underperforming relative to their salary and age (overvalued). Hamilton is not included in this table, having been excluded from the underlying model as a gross outlier, as noted above; this exclusion is itself informative, since Hamilton's very high salary paired with a low head-to-head win percentage represents one of the most extreme overvaluation cases in the sample by any raw comparison, even though the fitted model could not stably accommodate this observation. Table 3 presents this measure for all 19 drivers retained in the robust-regression sample, ranked from most undervalued to most overvalued.

Discussion

These results suggest that the salary level of drivers relates to on-track performance in F1, but that this relationship cannot be observed by the typical OLS regression in a small, highly skewed sample. Conventional regression yielded no significant results for salary in any of the specifications tested and the overall model accounted for little variance in performance. This result, however, was largely due to a few highly outlying observations with high values on the predictor, which have disproportionate influence on the regression line, a limitation of OLS regression on small samples with highly skewed predictor distributions.

If this outlier influence was removed by using robust regression, which down-weights extreme observations, then the salary became a statistically significant predictor of the head-to-head performance. This should be read as a sensitivity finding rather than definitive proof of an underlying causal relationship: one alternative estimator producing a significant result, in a single-season sample of fewer than twenty salary observations, indicates that the null OLS result was not the only reasonable conclusion available from this data, but it does not by itself establish that pay causally drives performance or vice versa. Substantively, this also means that the best cases of over and undervaluation in F1 are clustered around the highest-paid

drivers as these extreme cases are the ones that have the most influence on, and were initially hidden in, the overall pay-performance relationship.

Table 3

Driver-Level Over/Undervaluation, Ranked by Model Residual

Driver	Salary ($M)	Actual H2H %	Predicted H2H %	Residual
George Russell	15.00	85.0	54.7	+30.3 (most undervalued)
Isack Hadjar	0.75	63.2	33.2	+30.0
Oliver Bearman	1.00	55.0	32.5	+22.5
Alexander Albon	8.00	64.7	48.9	+15.8
Pierre Gasly	10.00	66.7	51.1	+15.6
Gabriel Bortoleto	2.00	46.7	34.6	+12.1
Charles Leclerc	34.00	85.7	76.6	+9.1
Fernando Alonso	20.00	82.4	76.6	+5.8
Oscar Piastri	6.00	45.5	41.9	+3.6
Liam Lawson	1.00	36.8	35.4	+1.4
Esteban Ocon	7.00	45.0	47.8	-2.8
Nico Hulkenberg	7.00	53.3	56.5	-3.2
Lando Norris	20.00	54.5	59.2	-4.7
Max Verstappen	65.00	100.0	110.7	-10.7
Kimi Antonelli	2.00	15.0	32.7	-17.7
Carlos Sainz	10.00	35.3	53.1	-17.8
Lance Stroll	3.00	17.6	41.5	-23.9
Jack Doohan	0.75	0.0	34.2	-34.2
Yuki Tsunoda	2.00	0.0	38.5	-38.5 (most overvalued)

Note. Predicted H2H % is each driver's fitted value from the robust regression model (salary and age as predictors, N = 19). Residual = Actual H2H % − Predicted H2H %. Positive residuals indicate performance above what salary and age predict (undervalued); negative residuals indicate performance below what salary and age predict (overvalued). Hamilton is excluded from this table; see discussion above.

The findings are consistent with the larger sports economics literature, which has consistently discovered that pay-performance relationships in professional sport are real, but have been hard to isolate in a clean fashion (Scully, 1974; Rosen, 1981). The present study builds on this literature by showing that a modest, one-season sample can show a meaningful pay-performance relationship after proper statistical corrections are made for the size of the sample and the impact of outliers.

Limitations

The findings of this study are limited in generalizability due to the small sample size (N = 20-21), the fact that all salary information was estimated rather than verified, and the exclusion of one driver due to missing salary information, and the single season nature of the study.

In addition, drivers from the same team are not statistically independent observations, since each head-to-head win percentage is mechanically determined by the paired outcome between two teammates; a driver-level regression such as the one used here may therefore understate the true dependence in the data. A team-pair or matched-difference analysis would address this limitation directly and is a priority for future work.

Conclusion

To examine the correlation between Formula 1 driver salary and on-track performance, the study analyzed all 24 races and six sprints of the 2025 season, using teammate head-to-head win percentage as the metric. The results indicated that salary was not a statistically significant predictor of head-to-head performance under standard OLS assumptions, but was a statistically significant predictor when the influence of a few very-high-paid drivers was down-weighted using robust regression. This backs up the overall theory that there is a correlation between pay and performance in this sport, but the correlation is found in a few extreme, headline salary situations and not evenly distributed throughout the driver field. The study advances the literature by using an empirical, car-quality-controlled, teammate-based measure of performance, rather than raw results, in the relationship between driver value and the current literature on this topic is largely opinion-based and is a paddock discussion.

The results have implications for F1 teams and commentators, as it indicates that the value of a driver should be determined using outlier-aware statistical methods, like the residual-based ranking obtained in this study (Table 3), and not simply based on a single linear model and/or anecdote. However, the study has a number of limitations: it was conducted over one season only, the number of drivers was small (19-21) and salary data was incomplete for some drivers and may thus limit the generalizability of the findings. Future research is invited to explore this relationship in other seasons and other sports where equipment has a similar role to determine if the pattern observed here is sustained over time and across sports. Finally, the study contributes to the understanding of the relationship between driver pay and on-track performance in Formula 1 and provides a first defensible approach to considering driver value as an empirical rather than a taste question.

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